Valid Sudoku (Medium)

Problem Statement

Determine if a 9x9 Sudoku board is valid. A valid Sudoku board will hold the following conditions:

1. Each row must contain the digits 1-9 without repetition.
2. Each column must contain the digits 1-9 without repetition.
3. The 9 3x3 sub-boxes of the grid must also contain the digits 1-9 without repetition.

Note:

1. The Sudoku board could be partially filled, where empty cells are filled with the character '.'.
2. You need to validate only filled cells.

Example 1:

- **Input:**

```
[["5","3",".",".","7",".",".",".","."]
,["6",".",".","1","9","5",".",".","."]
,[".","9","8",".",".",".",".","6","."]
,["8",".",".",".","6",".",".",".","3"]
,["4",".",".","8",".","3",".",".","1"]
,["7",".",".",".","2",".",".",".","6"]
,[".","6",".",".",".",".","2","8","."]
,[".",".",".","4","1","9",".",".","5"]
,[".",".",".",".","8",".",".","7","9"]]
```

- **Expected Output:** `true`
- **Justification:** This Sudoku board is valid as it adheres to the rules of no repetition in each row, each column, and each 3x3 sub-box.

Example 2:

- **Input:**

```
[["8","3",".",".","7",".",".",".","."]
,["6",".",".","1","9","5",".",".","."]
,[".","9","8",".",".",".",".","6","."]
,["8",".",".",".","6",".",".",".","3"]
,["4",".",".","8",".","3",".",".","1"]
,["7",".",".",".","2",".",".",".","6"]
,[".","6",".",".",".",".","2","8","."]
,[".",".",".","4","1","9",".",".","5"]
,[".",".",".",".","8",".",".","7","9"]]
```

- **Expected Output:** `false`
- **Justification:** The first and fourth rows both contain the number '8', violating the Sudoku rules.

Example 3:

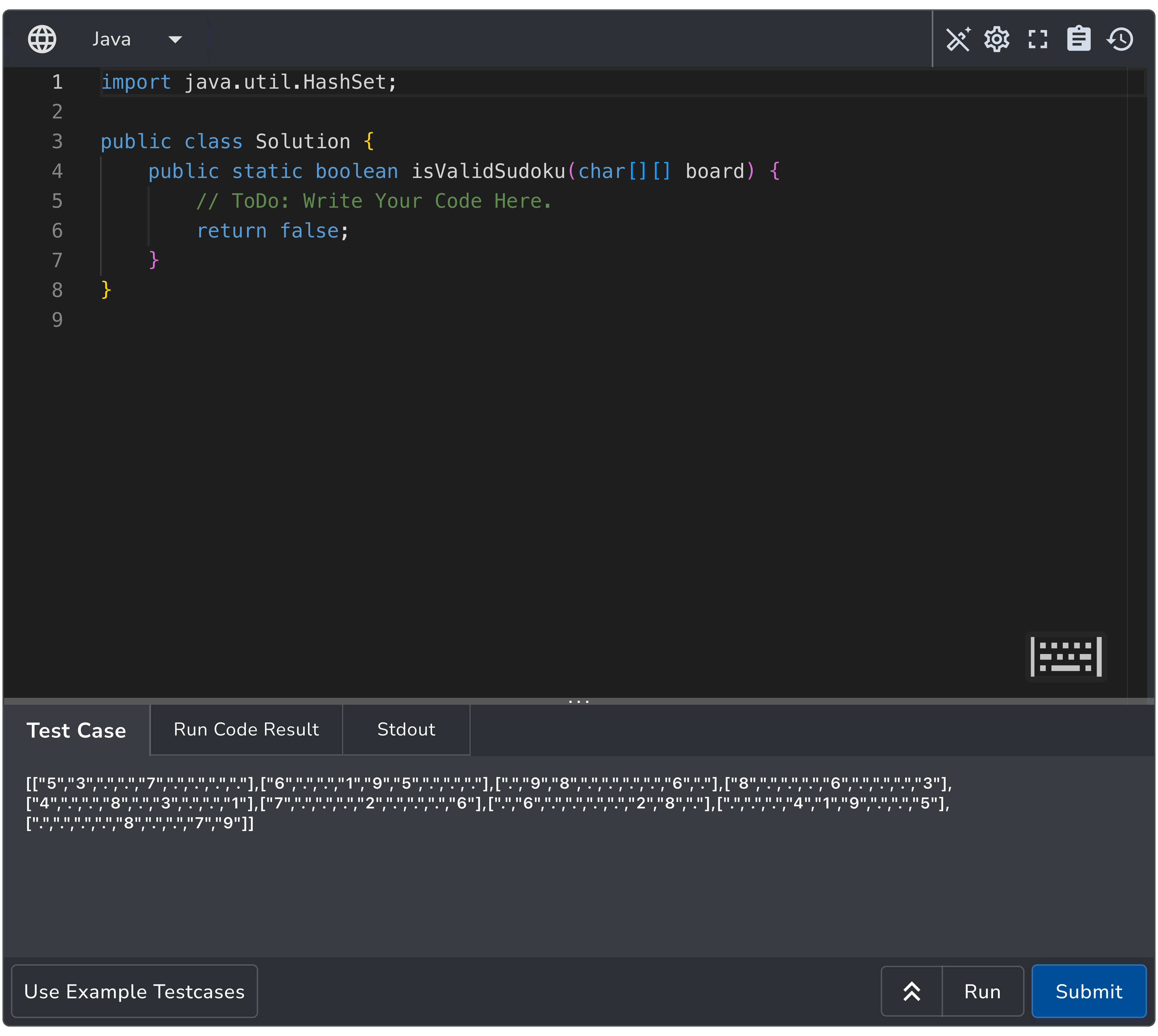
- **Input:**

```
[[".",".","4",".",".",".","6","3","."]
,[".",".",".",".",".",".",".",".","."]
,["5",".",".",".",".",".",".","9","."]
,[".",".",".","5","6",".",".",".","."]
,["4",".","3",".",".",".",".",".","1"]
,[".",".",".","7",".",".",".",".","."]
,[".",".",".","5",".",".",".",".","."]
,[".",".",".",".",".",".",".",".","."]
,[".",".",".",".",".",".",".",".","."]]
```

- **Expected Output:** `false`
- **Justification:** The fourth column contains the number '5' two times, violating the Sudoku rules.

Try it yourself

Try solving this question here:



```java
import java.util.HashSet;

public class Solution {
    public static boolean isValidSudoku(char[][] board) {
        // ToDo: Write Your Code Here.
        return false;
    }
}
```

```
[["5","3",".",".","7",".",".",".","."],["6",".",".","1","9","5",".",".","."],[".","9","8",".",".",".",".","6","."],["8",".",".",".","6",".",".",".","3"],
["4",".",".","8",".","3",".",".","1"],["7",".",".",".","2",".",".",".","6"],[".","6",".",".",".",".","2","8","."],[".",".",".","4","1","9",".",".","5"],
[".",".",".",".","8",".",".","7","9"]]
```

← Previous
Solution: Decode String

Next →
Solution: Valid Sudoku

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